Run report: m = 3 symmetric-S_4 solvers

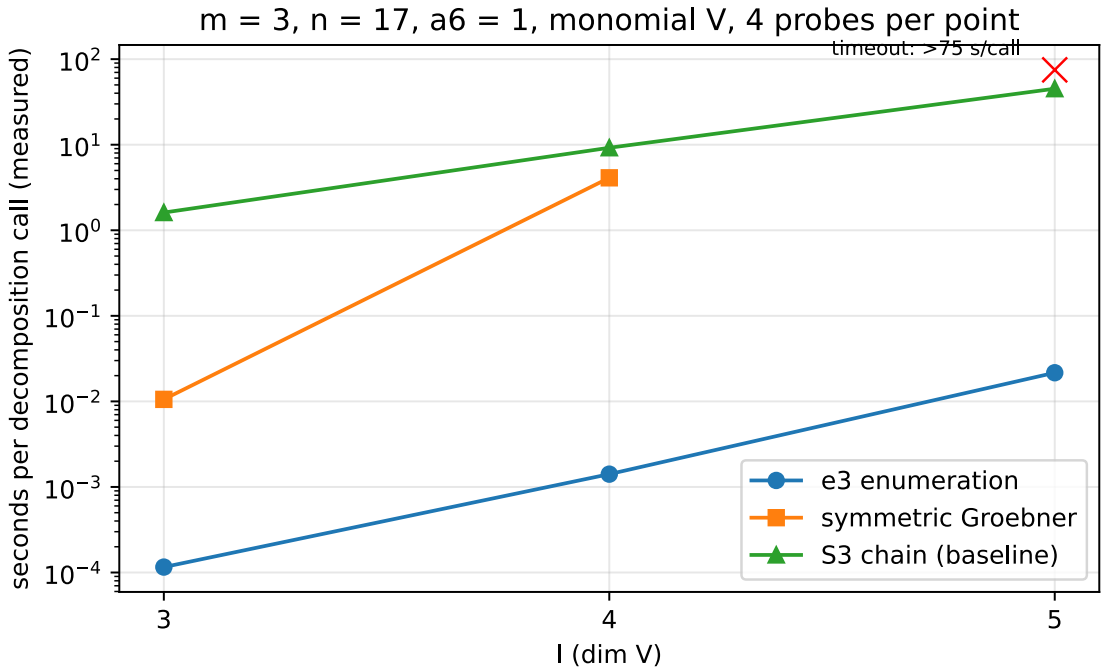
2026-10-11, UTC 19:55 · base 4b6e09b78 · 4-vCPU shared cloud container (wall clock indicative)

Runs

1. Full solves: e_3 enumeration, n = 17 (l = 5) and n = 19 (l = 6), 4 curves × {monomial, geometric g1}, 50 yield probes then closure. Records: `experiments/koblitz_m3/full_solves_e3.jsonl`.
2. Solver comparison at n = 17, a_6 = 1, monomial V, 4 probes, l = 3, 4, 5. Records: `experiments/koblitz_m3/solver_compare_n17.jsonl`.

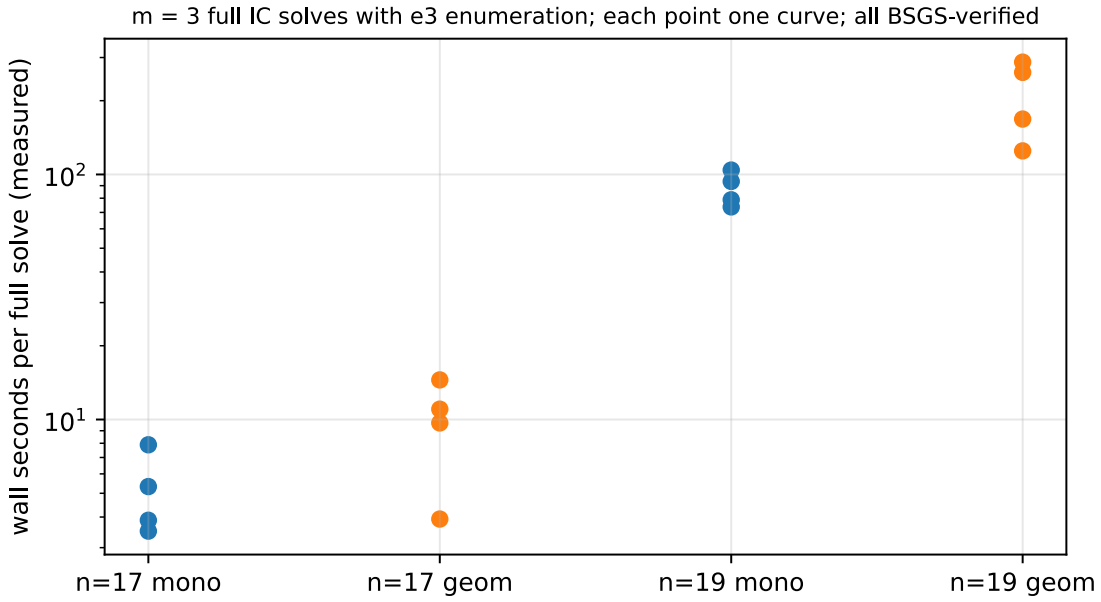
Per-call seconds (measured), n = 17

l	e_3 enumeration	symmetric Gröbner	S_3 chain
3	0.000116	0.0106	1.61
4	0.00141	4.10	9.21
5	0.0216	timeout (>300 s / 4 probes)	45.2



Full solves (measured wall seconds; all 16 verified against BSGS)

	monomial V	geometric V (g1)
n = 17, l = 5	3.5, 5.3, 7.9, 3.9	3.9, 9.7, 11.0, 14.5
n = 19, l = 6	93.6, 104.2, 73.7, 78.8	287.3, 260.9, 168.3, 124.8



Failures

One cell timed out (symmetric Gröbner, l = 5): a cost, not a mathematical result.